ENSTA Paris, **ENSAE**

Master SFA

(Elements of Stochastic Calculus-First Part, FQ307-OMI302, FR) 2020-21

EPREUVE DU 18 JANVIER 2021

Let $(\Omega, \mathcal{F}, P)$ be a complete probability space. Let $T > 0$. Let $(W_t)_{t\geq 0}$ be a standard $\mathbb{R}$-valued $(\mathcal{F}_t)_{t\geq 0}$-Brownian motion, where $(\mathcal{F}_t)_{t\geq 0}$ is a filtration fulfilling the usual conditions. All the questions are independent.

1. Let $n \in \mathbb{N}$. We define

$$M_t = \int_0^t (W_s)^n dW_s, \quad t \in [0, T].$$

Show that M is an $(\mathcal{F}_t)$-martingale, such that $M_t \in L^p$, for every $p \in [1, \infty[$.

2. Let ξ being an $\mathcal{F}_T$-measurable r.v. Let $a_0 \in \mathbb{R}$ and Y be a progressively measurable process such that

$$\xi = a_0 + \int_0^T Y_s dW_s, \tag{1}$$

where previous integral is an Itô integral.

(a) Explain why, if ξ is square integrable, the representation (1) is always possible.

(b) Suppose now that

$$E\left(\left(\int_0^T Y_s^2 ds\right)^{\frac{1}{2}}\right) < \infty.$$

Show that $\xi \in L^1(\Omega)$.

3. Let $X^{(i)}, i = 1, 2$, be two square integrable martingales such that

$$E\left((X_t^{(1)})^2\right) \leq E\left((X_t^{(2)})^2\right), \quad t \geq 0.$$

We set

$$\bar{X}_t^{(i)} := \int_0^t X_u^{(i)} du, i = 1, 2.$$

(a) Calculate $[\bar{X}_t^{(i)}, \bar{X}_t^{(i)}], i = 1, 2$.

(b) Show that, for $i = 1, 2$,

$$E\left(\int_{[0,T]^2} |X_t^{(i)} X_u^{(i)}| du dt\right) < \infty.$$

(c) Show that $E\left((\bar{X}_t^{(1)})^2\right) \leq E\left((\bar{X}_t^{(2)})^2\right)$, $t \in [0, T]$.

4. Let ξ be a Bernoulli($\frac{1}{2}$) r.v. independent of W. In particular we have $P\{\xi = 0\} = \frac{1}{2} = P\{\xi = 1\}$. We define the process $\mathcal{W}_t = (2\xi - 1)W_t, t > 0$.
   (a) Show that the law of $\mathcal{W}_t$ is $N(0, t)$.
   (b) Calculate the probability of the event $\{\xi W_t = 0\}$ for $t > 0$.
   (c) Is ξW_t a Gaussian r.v. ?
   (d) Show that the vector $(W_t, \mathcal{W}_t)$ is not a Gaussian vector.
   (e) Calculate the quadratic variation (if it exists) of $(\xi W_t)_{t\geq 0}$.
5. Let X be an Itô process of the type

$$X_t = M_t + \int_0^t b_s ds, \ t \in [0, T],$$

   with $M_t := \int_0^t a_s dW_s, \ t \in [0, T]$, where a and b are progressively measurable bounded processes. We define

$$N_t := \exp\left(-\int_0^t X_s dW_s - \int_0^t \frac{X_s^2}{2} ds\right).$$

   We admit without justification that there is a constant $c > 0$ such that

$$E(\exp(c \sup_{t\leq T} M_t^2)) < \infty.$$

   (a) Show that N is a martingale.
   (b) We set

$$\widetilde{W}_t = W_t + \int_0^t X_s ds, \ t \in [0, T].$$

   Give a probability Q, equivalent to P on $\mathcal{F}_T$, under which $\widetilde{W}$ is a Brownian motion.
   (c) Calculate $[\widetilde{W}]$.

Barême indicatif.

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